

Anomalous Thermoelectric Transport of Dirac Particles in Graphene

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We report a thermoelectric study of graphene in both zero and applied magnetic fields. As a direct consequence of the linear dispersion of massless particles, we find that the Seebeck coefficient S_{xx} diverges with $1/\sqrt{|n_{2D}|}$, where n_{2D} is the carrier density. We observe a very large Nernst signal S_{xy} ($\sim 50 \mu\text{V/K}$ at 8 T) at the Dirac point, and an oscillatory dependence of both S_{xx} and S_{xy} on n_{2D} at low temperatures. Our results underscore the anomalous thermoelectric transport in graphene, which may be used as a highly sensitive probe for impurity bands near the Dirac point.

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The unusual band structure of graphene gives rise to a host of intriguing phenomena in electrical transport properties that have been under extensive experimental investigations [1–6]. In solids, both charge and heat flows are simultaneously generated when an electrochemical potential or a temperature gradient is present, leading to additional effects. Fundamentally related to the electrical conductivity, other transport coefficients such as thermal conductivity and thermoelectric coefficients are also determined by the band structure and scattering mechanisms. Thermoelectric coefficients, in particular, involve the energy derivatives of the electrical transport counterparts such as the conductivity σ and the Hall angle Θ_H . The anomalies in the latter are very often amplified and cause markedly distinct features in the former near the Dirac point. Furthermore, in the regime where the Mott relation is applicable, the relationship between the measured electrical conductivity and the Seebeck coefficient reveals how the chemical potential depends on the gate voltage or carrier density, which is dictated by the energy dispersion. Therefore, the thermoelectric transport coefficients can offer unique information and are complementary to the electrical transport coefficients. A number of theoretical predictions have been made on transport coefficients other than electrical conductivities in graphene [7–9] which to date remain experimentally unexplored.

Single-layer graphene sheets are mechanically exfoliated onto degenerately doped silicon substrates that are covered with 300 nm of silicon dioxide. After locating suitable graphene sheets, we perform standard electron-beam lithography to attach electrodes in Hall-bar geometry. The electrodes consist of 7 nm of Cr and 100 nm of Au, and also serve as local thermometers. A microfabricated heater located on the right of Fig. 1(a) generates nearly parallel constant temperature contours along the graphene sample. The thermal voltage generated is measured across the two parallel Cr/Au electrodes $\sim 20 \mu\text{m}$ apart. These also double as local thermometers whose resistance is measured by the four-point method. A temperature difference of $\sim 10 \text{ mK}$ between the two Cr/Au wires can be readily measured for temperature $T > 10 \text{ K}$. An additional

pair of Cr/Au leads is used for transverse (Hall or Nernst) voltage measurements. All measurements were carried out in a cryostat with T ranging from 1.5 to 300 K and magnetic field B up to 8 T. In this work, the results are from two representative devices (#1 and #2) out of approximately two dozen fabricated devices. They are single-layer graphene as determined from optical images, and are often corroborated by the well-defined half-integer quantum Hall effect at low temperature. The carrier mobility μ_c is typically $\sim 3000 \text{ cm}^2/\text{Vs}$.

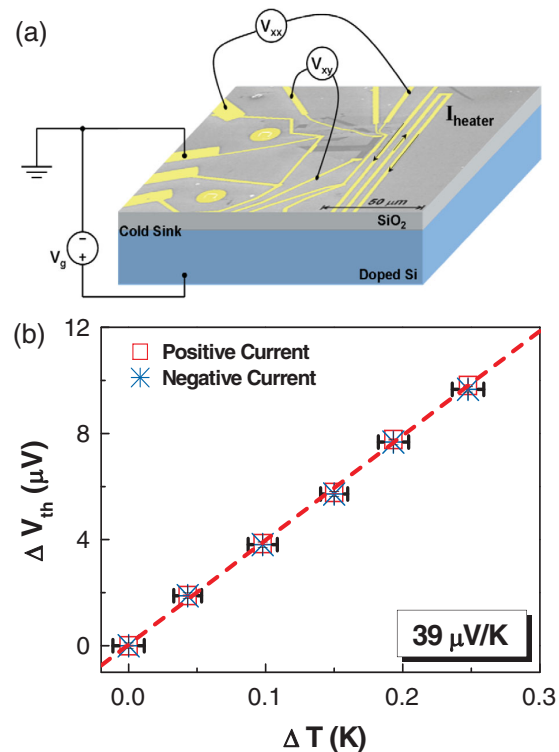


FIG. 1 (color). (a) SEM image and circuit schematic of a graphene device for thermoelectric measurements. (b) ΔT vs thermovoltage change ΔV_{th} for a series of heater power steps at 255 K and zero gate voltage. The linear fit of this curve gives the thermopower of $39 \mu\text{V/K}$.

石墨烯中狄拉克粒子的反常热电输运

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我们报告了石墨烯在零磁场和外加磁场中的热电特性研究。作为无质量粒子线性色散的直接结果, 我们发现塞贝克系数 S_{xx} 随 $1 = \sqrt{1 - 2Dj}$, 其中 n_{2D} 为载流子密度。我们在狄拉克点观察到非常大的能斯特信号 S_{xy} (在 8 T 时约为 $50 \mu V = K$), 并且在低温下 S_{xx} 和 S_{xy} 均随 n_{2D} 呈现振荡依赖性。我们的结果强调了石墨烯中异常的热电输运现象, 这可能被用作探测狄拉克点附近杂质能带的高灵敏度探针。

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石墨烯独特的能带结构在电输运特性中引发了一系列引人入胜的现象, 这些现象已得到广泛实验研究[1–6]。在固体中, 当存在电化学势或温度梯度时, 电荷流与热流会同时产生, 从而导致额外效应。与电导率密切相关的是, 其他输运系数 (如热导率和热电系数) 也由能带结构和散射机制决定。特别是热电系数, 涉及电输运对应量 (如电导率 σ 和霍尔角 Θ_H) 的能量导数。后者的异常现象常被放大, 导致前者在狄拉克点附近呈现显著不同的特征。此外, 在莫特关系适用的范围内, 测量电导率与塞贝克系数之间的关系揭示了化学势如何依赖于栅极电压或载流子密度, 而这些参数由能量色散决定。因此, 热电输运系数可以提供独特信息, 与电输运系数互补。目前已有若干理论预测了石墨烯中除电导率外的其他输运系数[7–9], 但这些预测至今仍未得到实验验证。

单层石墨烯片通过机械剥离法沉积在退化掺杂的硅基底上, 该基底表面覆盖有 300 纳米厚的二氧化硅层。在定位到合适的石墨烯片后, 我们采用标准电子束光刻技术以霍尔条形几何结构固定电极。这些电极由 7 纳米厚的铬层和 100 纳米厚的金层构成, 同时兼具局部温度计功能。如图 1(a) 右侧所示的微加工加热器, 可在石墨烯样品表面形成近乎平行的恒温等高线。通过间距约 $20 \mu m$ 的两根平行铬金电极测量产生的热电压, 这些电极同时也作为局部温度计使用, 其电阻值通过四点法进行测量。当温度 $T > 10 K$ 时, 两根铬金导线间的温差可轻松测量至约 10 mK。此外

一对 Cr=Au 引线用于横向 (霍尔或能斯特) 电压测量。所有测量均在低温恒温器中进行, 温度范围为 1.5 至 300 K, 磁场强度最高达 8 T。本研究结果来自约二十个制备器件中的两个代表性样品 (#1 和 #2)。经光学图像分析确认, 这些器件为单层石墨烯, 其低温下明确的半整数量子霍尔效应通常可作为佐证。载流子迁移率 μ_c 通常约为 $3000 \text{ cm}^2 = V s$ 。

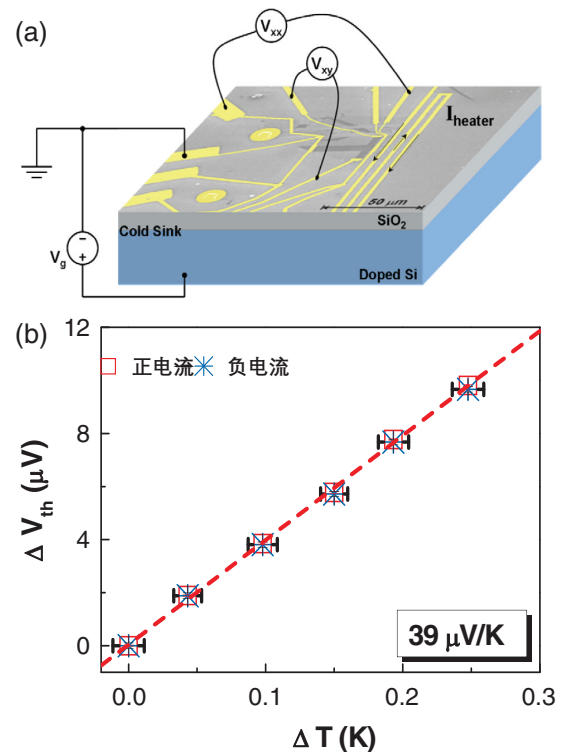


图1 (彩色)。(a)用于热电测量的石墨烯器件的SEM图像和电路示意图。(b)在255K和零栅极电压下, 一系列加热功率阶跃的 ΔT 与热电压变化 ΔV_{th} 的关系。该曲线的线性拟合得到热电势为 $39 \mu V = K$ 。

We generate a temperature gradient and measure both the temperature difference ΔT and thermal voltage change ΔV_{th} . Fitting a straight line to the ΔV_{th} vs ΔT data, we extract the Seebeck coefficient $S_{xx} = -\frac{\Delta V_{th}}{\Delta T}$ from the slope [Fig. 1(b)] [10]. At zero magnetic field, σ exhibits the characteristic minimum at $V_g \sim V_D$, the Dirac point. Figure 2(a) shows V_{th} as a function of V_g for three temperatures. V_{th} undergoes a sign change at the Dirac point $V_g = V_D = 10$ V, indicating the carrier type changes from hole to electron as $V_g - V_D$ is swept from negative to positive. V_{th} has a finite slope near V_D over a 20 V range in V_g which corresponds to $\sim \pm 100$ meV change in chemical potential μ measured from the Dirac point. This region coincides with the minimum in σ , where charged impurities modify the conductivity [11–14]. As V_g is further away from V_D on both sides, the magnitude of V_{th} decreases, scaling approximately with $1/\sqrt{|V_g - V_D|}$ [dashed line in Fig. 2(a)]. This V_g dependence is more noticeable in the linear dependence of $1/V_{th}^2$ on V_g [Fig. 2(b)]. The dashed lines are the power-law fits with exponent ~ 0.95 and cross zero in the vicinity of the Dirac point from both sides, indicating a diverging behavior of S_{xx} . Note that near the Dirac point, V_{th} crosses zero, and the $1/\sqrt{|V_g - V_D|}$ dependence breaks down, as denoted by the hatched region. For comparison, the same V_{th} data are also plotted as $1/|V_{th}|$ vs V_g in Fig. 2(c), and the straight lines are drawn in the linear region. Clearly, the $1/V_{th}^2$ plot shows a better linear relationship with V_g over the whole range. In addition, $1/V_{th}^2$ extrapolates to zero at almost the same V_g for different temperatures, but $1/|V_{th}|$ does not.

The fact that $|V_{th}|$ or $|S_{xx}|$ diverges as $1/\sqrt{|V_g - V_D|}$ is actually a direct manifestation of the linear dispersion of the Dirac particles in graphene. Let us assume $\sigma \sim |\mu|^\alpha$, which is sufficiently general to include both dirty ($\alpha \sim 2$) and clean ($\alpha \sim 1$) limits [12,13]. For degenerate electron systems, we expect the Mott relation $S_{xx} = -\frac{\pi^2 k_B^2 T}{3e} \frac{\partial \ln \sigma(\mu)}{\partial \mu}$ to hold, yielding $S_{xx} \sim -\frac{1}{\mu}$ for highly doped regimes. On the other hand, for a 2D system with a linear dispersion relation, then we expect $\mu = \hbar v_F \sqrt{n_{2D} \pi} \propto \pm \sqrt{|V_g - V_D|}$, where the $+$ ($-$) sign corresponds to the electron- (hole-) doped regime, and v_F is the Fermi velocity. Combining these relations, we have $S_{xx} \sim \frac{-\text{sgn}(\mu)}{\sqrt{|V_g - V_D|}}$. This is in contrast to the ordinary 2D electron systems with a quadratic dispersion relation, in which $\mu \propto n_{2D}$, and hence $S_{xx} \sim \frac{-1}{V_g - V_D}$. From this diverging behavior of S_{xx} , we can conclude that the dispersion relation is linear rather than quadratic, as expected for Dirac particles. It is worth noting that the exponent α is absorbed in the prefactor of S_{xx} and does not affect the functional dependence of S_{xx} , as is the case in σ . This makes the thermoelectric transport uniquely sensitive to the electronic band structure.

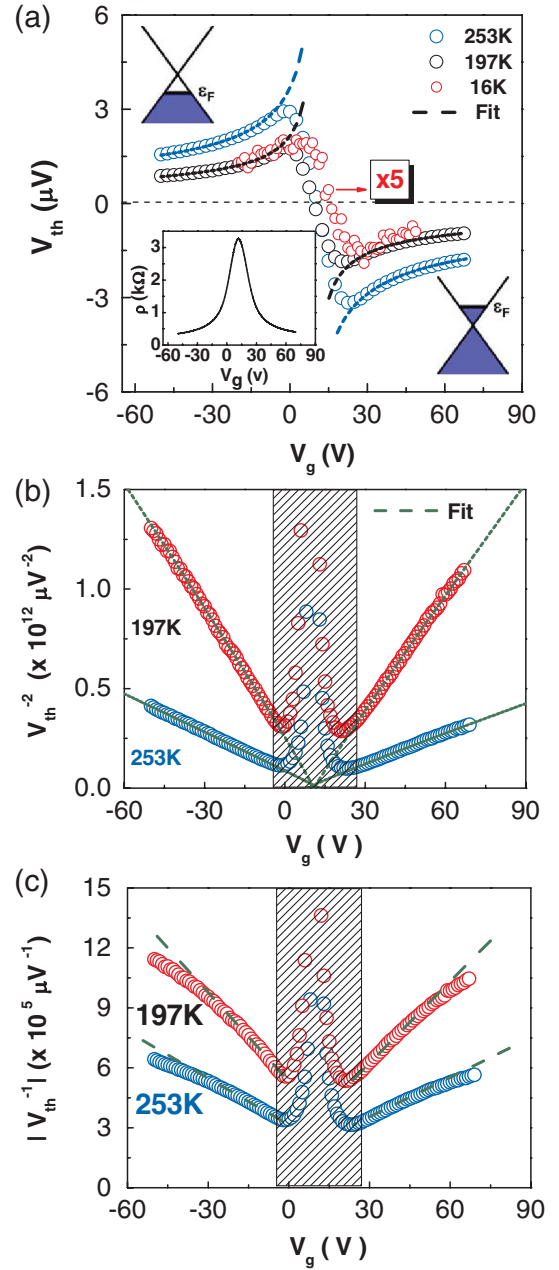


FIG. 2 (color). (a) V_{th} vs V_g for three different temperatures. The 16 K data (red circle) were multiplied by a factor of 5. The dashed lines are the fits described by $|S_{xx}| \sim 1/\sqrt{|V_g - V_D|}$. (b) $1/V_{th}^2$ vs V_g plot for the same data shown in (a). The shaded area is for $|V_g - V_D| < 10$ V. Green dashed lines are the best power-law fits with exponent ~ 0.95 . (c) $1/|V_{th}|$ vs V_g plot for the same data in (a). Green dashed lines are straight lines as guides to the eye.

Not every device shows the electron-hole symmetry shown in Fig. 2. Figure 3(a) displays S_{xx} vs V_g of a different device with $V_D \sim 33$ V for several values of T ranging from 11 to 255 K. Away from V_D on the hole side, S_{xx} decreases with decreasing V_g , similar to the behavior of the previous device. In contrast, S_{xx} stays flat on the electron side, indicating a strong electron-hole asymmetry as seen

我们生成温度梯度并测量温度差 ΔT 和热电压变化 ΔV_{th} 。通过将直线拟合到 ΔV_{th} 与 ΔT 数据，我们 [xFtiring t l t (hbe)] e[1 eb0] kActozeffi ero magnetic fi-el d, frσ exhi hbeitss otphee在 $V_g \sim V_D$ 处具有特征最小值，即狄拉克点。图2(a)展示了三种温度下 V_{th} 随 V_g 的变化关系。 V_{th} 在狄拉克点 $V_{g,hol}$ 处发生符号变化。 V_{th} 在 V_D 附近具有有限斜率，其变化范围在 V_g 的 20V 范围内，对应于从狄拉克点测量的化学势 μ 变化约 ± 100 meV。该区域与 σ 的最小值重合，带电杂质在此处改变电导率[11–14]。随着 V_g 在两侧与 V_D 的距离增加， V_{th} 的幅值减小，其变化量近似遵循 $1 = \sqrt{\dots} jV_g - \dots V_D j$ 的规律[图2(a)中的虚线]。这种 V_g 依赖性在 $1 = V_{th}^2$ 与 V_g 的线性关系中更为明显[图2(b)]。虚线为幂律拟合曲线，指数约为 0.95，且在狄拉克点附近两侧均穿过零点，表明 S_{xx} 呈现发散行为。需注意在狄拉克点附近， V_{th} 穿过零点，且 $1 = \sqrt{\dots} jV_g - \dots V_D j$ 的依赖关系失效，如图中阴影区域所示。为便于比较，图2(c)中还绘制了相同 V_{th} 数据的 $1 = |V_{th}|$ 与 V_g 关系图，其中直线段绘制于线性区域。显然， $1 = V_{th}^2$ 曲线在整个范围内与 V_g 的线性关系更优。此外当 $1 = V_{th}^2$ 在不同温度下几乎相同的 V_g 处外推至零，但 $1 = |V_{th}|$ 则不然。事实上， $|V_{th}|$ 或 $|S_{xx}|$ 会随着 $1 = \sqrt{\dots} jV_g - \dots V_D j$ 而发散。 V_D 实际上是石墨烯中狄拉克粒子线性色散的直接表现。假设 $\sigma \sim j\mu j^\alpha$ ，该假设足够通用，可同时涵盖脏态 ($\alpha \sim 2$) 和干净态 ($\alpha \sim 1$) 极限[12, 13]。对于简并电子系统，we expect the Mott relation $S_{xx} = \frac{\pi^2 k_B^2 T}{3e} \frac{\partial \ln \sigma(\mu)}{\partial \mu}$ 保持，产生 $S_{xx} \sim \frac{1}{\mu}$ 对于高掺杂区域。另一方面，对于具有线性色散关系的二维系统，我们预期 $\mu = @v_F \sqrt{n_{2D}} \pi \alpha \pm \sqrt{\dots} jV_g - \dots V_D j$ ，其中 + (-) 符号对应电子- (空穴-) 掺杂区域， v_F 为费米速度。结合这些关系式，我们得到 $S_{xx} \sim \frac{-\text{sgn}(\mu)}{\sqrt{|V_g - V_D|}}$ 这与具有二次色散关系的普通二维电子系统形成对比，在后者中 $\mu \propto n_{2D}$ ，因此 $S_{xx} \sim \frac{-1}{V_g - V_D}$ 从 S_{xx} 的这种发散行为中，我们可以得出结论：其色散关系是线性的而非二次方的，这与狄拉克粒子的预期相符。值得注意的是，指数 α 被吸收在 S_{xx} 的前因子中，并不会影响 S_{xx} 的泛函依赖性，这与 σ 的情况不同。这使得热电输运对电子能带结构具有独特的敏感性。

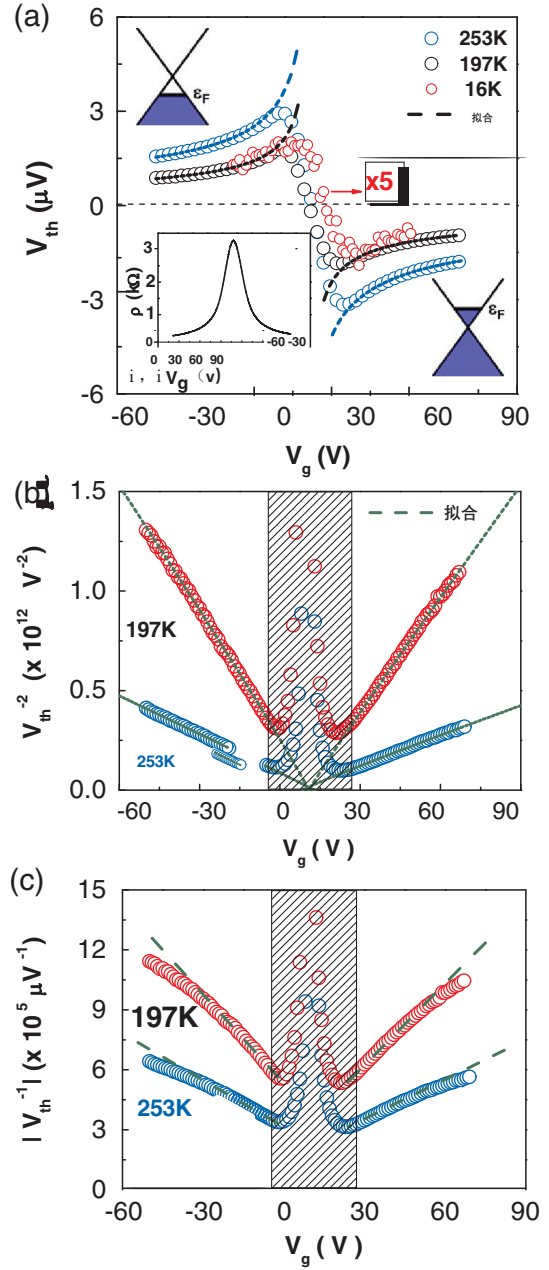


图2 (彩色)。(a) V_{th} 与 V_g 在三种不同温度下的关系。16 K 数据 (红色圆圈) 乘以 5 倍放大。虚线为 $|S_{xx}| \sim 1/\sqrt{|V_g - V_D|}$ 描述的拟合曲线。(b) $1/V_{th}^2$ 与 V_g 在同一数据下的关系图。阴影区域对应 $|V_g - V_D| < 10$ V。绿色虚线为最佳幂律拟合，指数约为 0.95。(c) $1/|V_{th}|$ 与 V_g 在同一数据下的关系图。绿色虚线为视觉引导的直线。

并非所有器件都表现出图2所示的电子-空穴对称性。图3(a)展示了另一器件的 S_{xx} 随 V_g 的变化关系，该器件的 V_D 约为 33 V，测试温度范围为 11 至 255 K。在远离 V_D 的空穴侧， S_{xx} 随 V_g 降低而减小，与前一器件的行为相似。相比之下， S_{xx} 在电子侧保持平坦，表明存在显著的电子-空穴不对称性。

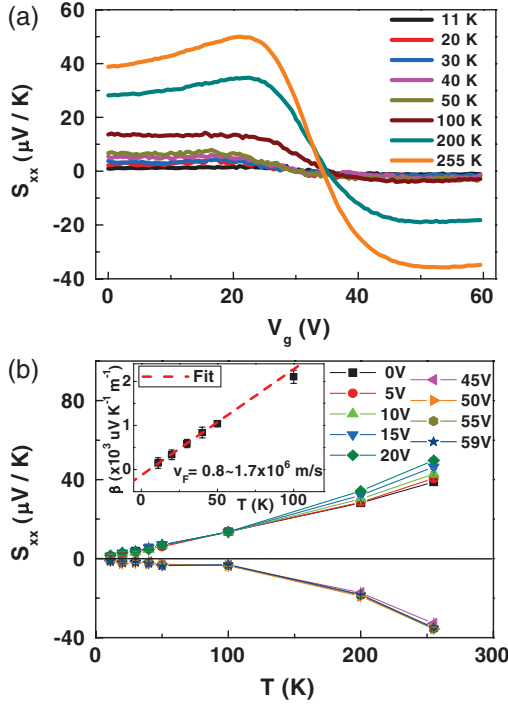


FIG. 3 (color). (a) V_g dependence of longitudinal Seebeck coefficient S_{xx} at different temperatures (11–255 K) and zero magnetic field. (b) T dependence of S_{xx} at different gate voltages. The inset is the T dependence of $\beta = S_{xx}\sqrt{n_{2D}}$ at $V_g = 0$ V for low temperatures. The slope of the linear fit is proportional to α/v_F .

in σ by others [12]. Near V_D , we observe a broad transition region in S_{xx} connecting the electron- to hole-doped regimes. Furthermore, S_{xx} follows different T dependence for different V_g [in Fig. 3(b)]. Near V_D , the magnitude of S_{xx} is close to zero. Away from V_D on the hole side, e.g., at $V_g = 0$ V or ~ 33 V left of V_D , S_{xx} is nearly a straight line for the whole temperature range. As V_g approaches V_D from the hole side, S_{xx} begins to deviate from the linear T dependence at progressively lower temperatures. On the electron side, however, even at $V_g = 60$ V (or ~ 30 V right of V_D), S_{xx} remains nonlinear in T except at very low temperatures.

The departure from the linear T dependence is an indication of the potential breakdown of the Mott relation. For this device, when $|V_g - V_D| = 30$ V, $|\mu|$ is about 160 meV measured from the Dirac point. It is reasonable to expect high-order corrections in the Sommerfeld expansion at relatively high temperature where the condition $|\mu| \gg k_B T$ fails. For graphene, another relevant energy scale is the bandwidth γ of impurity states [15,16] near the Dirac point. The Mott relation only holds if $\frac{\gamma}{k_B T} \gg 1$, which ensures σ to be a slow-varying function of energy over this band of impurity states [16]. In the impurity scattering model, this band can be highly asymmetric due to the finite scattering potential. Here, we attribute the departure from the linear T dependence on the electron

side to the asymmetric nature of the band of impurity states. For this reason, we only focus on the relatively low- T region on the hole side where the Mott relation apparently holds. Since S_{xx} is proportional to αT , and inversely proportional to μ or $v_F \sqrt{n_{2D}}$, we plot $S_{xx}\sqrt{n_{2D}}$ (called β) vs T in the inset of Fig. 3(b). Extracted from the slope, v_F ranges from 0.8 to 1.7×10^6 m/s depending on the value of α (from 1 to 2), which is in good agreement with the values obtained by others [17]. In relating V_g to n_{2D} for above estimations, we use $n_{2D} = \frac{C_g V_g}{e} + \bar{n} = \frac{C_g}{e}(V_g - V_D)$, where C_g is the capacitance per unit area and $\bar{n}$ is the induced density by charged impurities at the Dirac point. A value of $C_g = 103$ aF/ μm^2 is determined from our Hall data.

In a magnetic field, carriers diffusing under ∇T experience the Lorentz force, resulting in a nonzero transverse voltage V_y . The transverse effect or the Nernst effect is measured by $S_{xy} = -\frac{E_y}{|\nabla T|} = \frac{\Delta V_y}{\Delta T_x}$. In nonmagnetic metals, S_{xy} is negligibly small (~ 10 nV/K per tesla) [18]. In ferromagnets, spin-orbit coupling can lead to a large spontaneous Nernst signal [19]. Here, we observe an exceedingly large Nernst peak (~ 50 $\mu\text{V/K}$ at 8 T) at the Dirac point [Fig. 4(a)], and we attribute it to the unique band structure of graphene. In classical transport, the Mott relation takes the following form [7,20]: $S_{xy} = -\frac{\pi^2}{3} \frac{k_B^2 T}{e} \times \left(\frac{\partial \Theta_H}{\partial \epsilon}\right)_\mu = \frac{\pi^2 k_B^2 T B}{3} \frac{\partial}{\partial \mu} \left[\frac{\tau}{m^*} \text{sgn}(\mu)\right]$. S_{xy} is directly proportional to the energy derivative of the Hall angle Θ_H or inversely proportional to the cyclotron mass m^* . For massless particles, the vanishing cyclotron mass can indeed lead to a

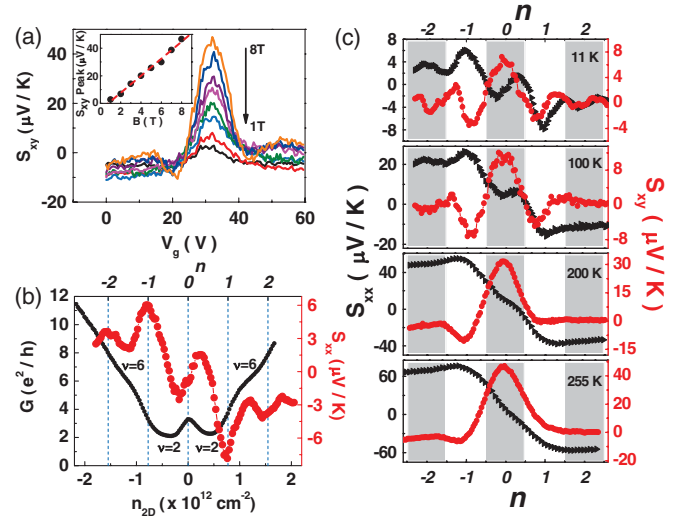


FIG. 4 (color). (a) V_g dependence of Nernst signal S_{xy} at 160 K with different magnetic fields (1–8 T). Inset: B dependence of S_{xy} at $V_g = V_D$, and the red line is a linear fit. (b) Two-terminal conductance G and thermopower S_{xx} vs carrier density n_{2D} at $T = 11$ K and $B = 8$ T. The corresponding Landau level index n is shown on the top axis. (c) S_{xx} (black triangle) and S_{xy} (red circle) vs Landau level index n for four different temperatures at $B = 8$ T.

diverging behavior in S_{xy} . In graphene devices, however, the anomaly is diminished by the impurity states near the Dirac point. Recall that the Mott relation breaks down in this region. Here, we estimate the magnitude of S_{xy} at the Dirac point both from Θ_H outside this region where the Mott relation holds and from γ . Since we have $\Theta_H = -\mu_c B \text{sgn}(\mu)$ (μ_c : carrier mobility), we obtain $\Delta\Theta_H \sim 2.2$ with an 8 T magnetic field at 255 K. This change in Θ_H occurs over $\gamma \sim 204$ meV as estimated from the width of the conductance minimum, yielding $S_{xy} \sim 68$ $\mu\text{V/K}$. This is in very good agreement with the experimentally observed peak value (~ 50 $\mu\text{V/K}$). Additionally, Θ_H is directly proportional to B , which indicates a linear B -field dependence in S_{xy} , with an estimated slope of ~ 5.4 $\mu\text{V/K}^*\text{T}$ at 160 K. Indeed, the linear B dependence of S_{xy} is observed [Fig. 4(a)], and the slope of the straight line is ~ 6 $\mu\text{V/K}^*\text{T}$. Similar to S_{xx} whose diverging behavior is greatly modified by the disorders, the anomaly in S_{xy} depends on the carrier mobility as well as γ . We expect to see more pronounced anomalous behavior in both S_{xx} and S_{xy} in cleaner samples.

At low temperatures and $B = 8$ T, the device conductance exhibits clear quantum Hall plateaus as V_g is varied. In this regime, we observe oscillations in S_{xx} [Figs. 4(b) and 4(c)] that are reminiscent of the Shubnikov–de Hass oscillations in ρ_{xx} [2,3], and the side peaks and dips in S_{xy} that correlate with the oscillatory structures in S_{xx} . At $T = 11$ K, S_{xx} shows peaks (dips) as μ is inside the broadened Landau levels (LL) on the hole (electron) side. These peaks (dips) correspond to the LL indices $n = 1$ and $n = 2$ for holes (electrons). S_{xy} also changes sign at these fillings. It is also worth noting that S_{xx} crosses zero at the Dirac point (in the lowest LL), accompanied by an additional small dip (peak) on the hole (electron) side. The origin of this feature is unknown, but it could reveal some peculiarities of the zeroth LL at high magnetic fields. In conventional 2D electron systems, the observed S_{xx} peaks at the LL's are consistent with the calculations in the integer quantum Hall regime [21]. In graphene samples, the $n = 1$ and $n = 2$ peaks in S_{xx} on both electron and hole sides are also expected. However, we do not observe vanishing S_{xx} as μ is located between the two adjacent LL's. The nonvanishing S_{xx} was previously attributed to the activated behavior in ordinary 2D electron systems. In our samples, the relatively large magnitude of S_{xx} between the LL's may be caused by the broadened LL's due to disorders. We expect to see $S_{xx} \rightarrow 0$ at low temperatures and the predicted activated behavior at high temperatures in cleaner samples.

As the temperature increases, the oscillations in S_{xx} and S_{xy} become weaker, although the overall magnitude of both S_{xx} and the central peak in S_{xy} increases [Fig. 4(c)]. As discussed earlier, the characteristic width of the Nernst peak is primarily determined by γ , which is greater than $k_B T$. The Nernst width remains nearly unchanged as a consequence.

In summary, the diverging behavior ($|S_{xx}| \sim 1/\sqrt{|n_{2D}|}$) of the Seebeck coefficient along with the exceedingly large Nernst peak at the Dirac point is characteristic of the massless particles in graphene. With disorders, these generic anomalies are somewhat masked near the Dirac point. However, the diverging behavior can be retrieved from those quantities as the chemical potential approaches the Dirac point. In higher mobility graphene samples, the anomalies are expected to be more drastically pronounced.

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